

Exam Date & Time: 12-Jan-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS -JANUARY 2024

SUBJECT: CSE 2123- DIGITAL SYSTEM DESIGN

DIGITAL SYSTEM DESIGN [CSE 2123]

Marks: 50

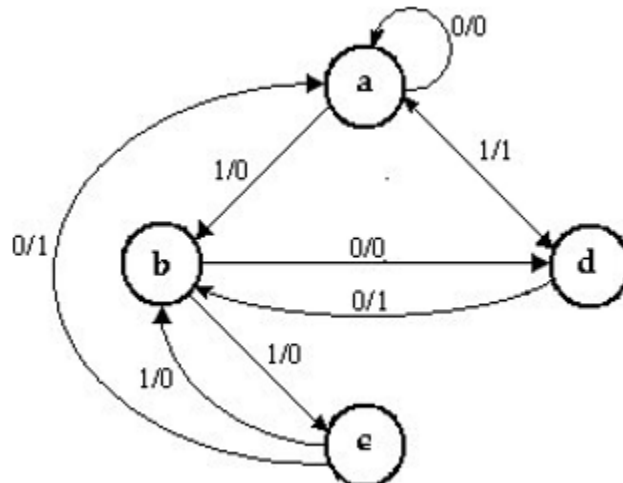
Duration: 180 mins.

A

Answer all the questions.

Missing data may be suitably assumed.

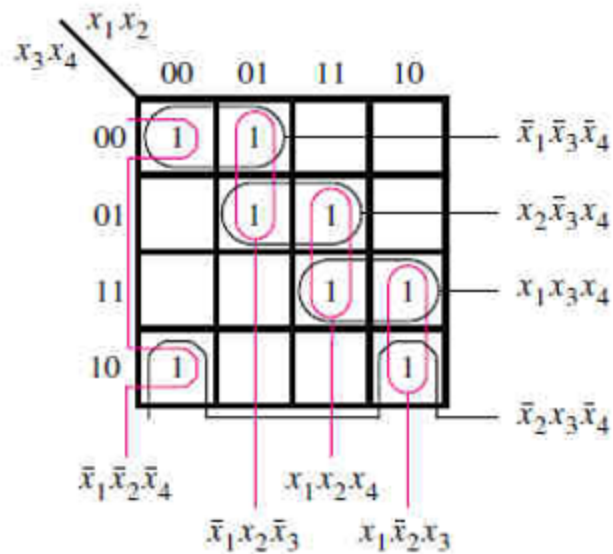
- 1A) For the following state diagram, derive the expressions required for the synchronous sequential circuit design using T flip-flops.



(5)

- 1B) Find the POS implementation for the following and compare the cost with the SOP form.

(3)



1C) Explain in detail the design of a basic RS latch using NOR gates. (2)

2A) Design a control unit of a simple processor to perform the operations mentioned in Table below. For the control unit,

- Develop the timing signals for the operations
- Tabulate the necessary control signals that needs to be asserted during the timings of a)
- Derive the expressions for all the control signals of b).

Operation	Function
I0: Load Rx, Data	$Rx \leftarrow \text{Data}$
I1: Move Rx, Ry	$Rx \leftarrow [Ry]$
I2: Swap Rx, Ry	$Rx \leftrightarrow Ry$
I3: Add Rx, Ry	$Rx \leftarrow Rx + Ry$
I4: Sub Rx, Ry	$Rx \leftarrow Rx - Ry$

(5)

Where,

- Rx and Ry are the registers from R0, R1, R2, R3. All these registers are connected to a common bus through a tri-state buffer.
- External data can be loaded into these registers with a tri-state buffer.
- Use accumulator (A) and base register (B) for temporary operations with Arithmetic unit.
- Each operation begins upon the assertion of signal W to 1
- Completion of each operation is indicated by asserting the signal Done = 1
- Use control signal $Add/\overline{Sub}$ to control the addition and subtraction operations.

2B) Implement the function $f(w_1, w_2, w_3) = \sum_m(1, 2, 3, 5, 6)$ can be formulated using a 3-to-8 decoder and an OR gate. (3)

2C) Explain how the restriction on the pulse width in the case of JK flip-flop can be solved by using master/slave JK flip-flop. (2)

3A) Design the synchronous counter with the following repeated binary sequence: 0, 1, 2, 3, (5)

4, 5, 6. Use JK flip-flops. Check if it is self-correcting or not.

- 3B) Analyze the following state table and perform the state reduction. Develop the state diagram for the reduced state table.

Present state	Next state		Output	
	x=0	x=1	x=0	x=1
a	a	b	0	0
b	c	h	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1
h	e	d	0	1

(3)

- 3C) Develop a state diagram for the FSM that meets the following specification:
The circuit has one input, w , and one output, z . The output z should be equal to 1 in the same clock cycle when the second occurrence of $w = 1$ is detected. Otherwise, the value of z is equal to 0.

(2)

What is the output z produced for the following input sequence?

w : 1 0 0 1 1 1 1 0 0 1 1

- 4A) Develop the ASM chart for the bit counting circuit which counts the number of 1's in a register. Explain the same. Justify how it is different from traditional flowchart.

(5)

- 4B) Construct a Johnson counter which generates 8 timing signals. Write the count sequence and AND gate expressions required for decoding the output.

(3)

- 4C) Construct CMOS NOR gate. Write the truth table and transistor states.

(2)

- 5A) Build the D-type positive edge-triggered Flip-flop using NAND gates. Explain its operation with the logic diagrams for all the cases of D when the clock is low. Also, predict the outputs of the flip-flop for each case assuming that the flip-flop is cleared previously.

(4)

- 5B) Design a hierarchical 40-bit carry look-ahead adder with ripple-carry between 8-bit blocks. Compute the total gate delay for the design mentioning all the intermediary gate delays.

(4)

- 5C) Construct a 5-bit ring counter using D flip-flops.

(2)

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